General Equilibrium Theory

A. Interdependence in the Economy

partial equilibrium (PE) approach deals with the particular segment of the economy in isolation of what was happening in other segments, under the assumption of ceteris paribus.

For example(1), individual achieves the goals of utility maximization attaining the condition

$$MRS_{XY} = P_X/P_Y$$; this condition holds under the assumption that income is given.

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However, income depends on the amount of labor and other factors of production that the consumer owns (or endowment) and on their prices (wage, rent, interest etc)

(2) A firm selects a combination of inputs that minimizes the cost of producing a specified level of output, under the condition that

$$MRTS_{LK} = P_L/P_K$$; this condition

holds under the assumption that factor prices, the state of technology and the prices of commodities are given. Firms takes the cost minimization decision in isolation of such factors as the demands for the products, which in turn are influenced by the level of employment, income and taste of consumers.

(3) A firm achieves its goal of profit maximization, under the condition that MR = MC.

This condition holds true regardless of the market structure within which the firm operates. That is in product market interaction among buyers and sellers in one market determine price and levels of output ignoring other markets.

(4) In factor markets, where firms and households are owners of productive resources interact with each other to deter prices and quantities of various factors employed, ignoring the interrelationship

between the various factor and commodity markets.

Thus in the partial equilibrium approach, the Marshallian methodology is regarded independently of the others.

However, a fundamental feature of any economic system in the interdependence among its constituent parts. product prices and factor prices are simultaneously determined. For example,

• consumers' demands for various goods and services depend on their tastes, and income.

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• consumers' income depend on ~~the amount~~ resource endowment and factor prices.

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• Factor prices depend on the demand and supply of the various inputs

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• The demand for factors by firms depends not only on the state of technology but also on the demands final goods they produce (derived demand ?)

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- The demands for these goods depend on consumers' income, which in turn depend on the demand for the factors of production.

These interdependence can be illustrated by the circular flow diagram under the assumption that

a) all production takes place in the business sector;

b) all factors of production are owned by the households;

c) all factors are fully employed;

d) all incomes are spent

The economic activity in the system takes the form of two flows between the consumer sector and the business sector: a real flow and a monetary flow.

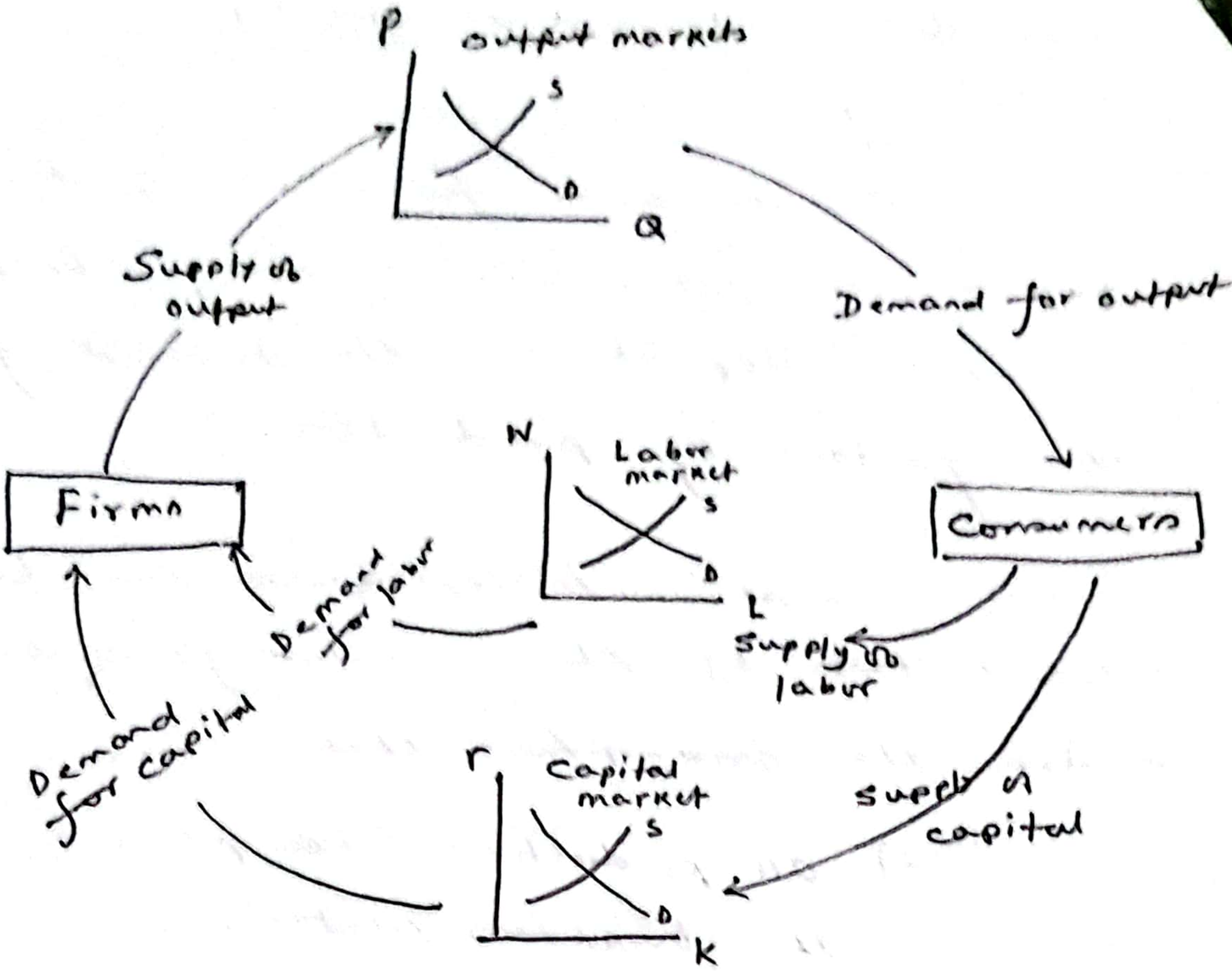

Fig: Circular flow diagram in a two sector economy

The real flow is the exchange of goods or the services of factors of production. The monetary flow is the real flow expressed in monetary terms. These two flows ~~which represent the transactions and the~~ move in opposite directions. It is through this constant interaction among the various economic agents that equilibrium in both output and input markets.

Economic system consists of millions of economic decision making units who are motivated by self interest. Each one pursues his own goal and strives for his own equilibrium independently of the others.

General equilibrium (GE) theory deals with the problem of whether the independent action by each decision maker leads to a position in which equilibrium is reached by all. A G.E. is defined as a state in which all markets and all decision-making units are in simultaneous equilibrium. The implication is that a G.E. exist if each market is cleared at a positive price, with each consumer maximizing satisfaction and each firm maximizing profit, so that there is neither excess demand nor excess supply in the market.

solution for million of unknown. The unknowns are the prices of all factors and all commodities and quantities purchased and sold (of factors and commodities) by each consumer and each producer. There are two types of equations

1) Behavioral equations describing the demand supply functions in all markets by all individuals.

2) Market clearing equations.

When does the solution is possible?

B. The Walrasian system

French Economist Leon Walras used a system of simultaneous equations to describe the interaction of individual sellers and buyers in all markets, and he maintained that all the relevant magnitudes (prices and quantities of all commodities and all factor services) can be determined simultaneous by the solution of this system.